difference between assume let consider suppose and these kinds of words mathematically

In mathematics, words like **assume, let, consider, suppose** look similar, but they are used in slightly different ways depending on the intention of the step in a proof or explanation.

Here is a clear, simple comparison.

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1) Let

Meaning: Introduce a new object or define something.

Use when: You are **creating or naming** something for the first time.

Example:

- Let $x = 5$
- Let n be a positive integer.
- Let $f(x) = x^2 + 1$

Idea:
You are **declaring** something, not guessing or assuming.

Shortcut meaning:
"From now on, we define this."

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2) Assume

Meaning: Accept something as true temporarily.

Use when:
You want to **use a condition** to prove something.

Example:

- Assume $x > 0$.
- Assume the function is continuous.
- Assume n is even.

Idea:
You are **temporarily treating something as true** to see what follows.

Common in proofs:

- Direct proofs
- Proof by contradiction

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3) Suppose

Meaning: Similar to *assume*, but slightly softer or more conversational.

Use when:
You want to explore a situation or start a logical argument.

Example:

- Suppose $x = 2$.
- Suppose the triangle is right-angled.
- Suppose n is divisible by 3.

Idea:
You are **imagining a case** to analyze it.

Difference from "assume":

- *Assume* sounds more formal and logical.
- *Suppose* sounds more explanatory or narrative.

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4) Consider

Meaning: Look at a specific object, case, or example.

Use when:
You want the reader to **focus on something specific**.

Example:

- Consider the function $f(x) = x^2$.
- Consider a triangle with sides 3, 4, and 5.
- Consider the equation $x^2 - 4 = 0$.

Idea:
You are saying:
"Let's look at this example or object."

Quick Comparison Table

Word	Main Purpose	Tone	Typical Use
Let	Define or introduce something	Neutral	Let $x = 5$
Assume	Treat something as true for a proof	Formal	Assume $x > 0$
Suppose	Imagine a case	Slightly informal	Suppose n is even
Consider	Focus on an object or example	Explanatory	Consider $f(x) = x^2$

Simple Example Showing All Together

Problem: Prove that the square of an even number is even.

Proof:

- **Let** n be an even number.
- **Assume** $n = 2k$ for some integer k.
- **Consider** n^2.
- **Suppose** we compute it:

$$n^2 = (2k)^2 = 4k^2 = 2(2k^2)$$

So n^2 is even.

Easy Memory Trick

- **Let** = define
- **Assume** = accept as true
- **Suppose** = imagine
- **Consider** = look at

If you want, I can give:

- Practice questions
- Examples from real proofs
- Differences in formal math writing vs school math.